Nice tests in the two-core model: where an inadmissible example could live

Notes on the question of whether all closed, convex, coordinatewise-lower exact-size tests are admissible when $\mathcal{C} = \{1, 2\}$ :: August 28, 2026

Throughout, $X \sim N_3(\mu, I_3)$, $\Lambda = \{\mu : \mu_3 \le \max\{\mu_1, \mu_2\}\}$, $\Lambda_0 = \Lambda \cap (-\infty, 0]^3$, and $\Lambda_1 = \{\mu \in \Lambda : \max\{\mu_1, \mu_2\} > 0\}$. Call an acceptance region A *nice* if it is closed, convex, coordinatewise lower, and $P_0(A) = 1 - \alpha$. Call a direction $a \ge 0$, $a \ne 0$, *feasible* if $a_3 \le \max\{a_1, a_2\}$ and *bad* otherwise. Note that every nonzero feasible direction belongs to Λ_1 (if $a_1 = a_2 = 0$ then feasibility forces $a_3 \le 0$, hence $a = 0$). For a nice A, lowerness implies that the size is attained at the origin, so any weak dominator ψ of $\phi_A := I\{x \notin A\}$ satisfies, by the zero-mass argument of Proposition 1 of the main note, $\mathrm{E}_0 h = 0$ with $h := \psi - \phi_A$, and by the first-moment argument there, $q := \mathrm{E}_0[Xh(X)] = 0$. Both arguments use only the model, not the shape of A, so they are available for every nice test.

1 Why examples are hard to find: three structural facts

The difficulty is not bad luck. The two-core model destroys every standard device for proving inadmissibility, for three related reasons.

First, the transfer cone is trivial. Section 2 of the main note shows that the only sample-space direction d with $\mu' d \ge 0$ for all $\mu \in \Lambda_1$ is $d = 0$. This is exactly the computation that makes Stein–Birnbaum-type *necessary* conditions for admissibility vacuous here: those conditions require the rejection region to be monotone in directions that raise every alternative likelihood ratio, and there are no such directions. No test can be disqualified on local-monotonicity grounds.

Second, the convex hull of the alternative directions is all of $\mathbf{R}^3$. For instance $(-1, -1, 0) = (1, -2, 0) + (-2, 1, 0)$ with both summands in Λ_1, and $(0, 0, \pm 1)$ arise similarly. This is why the first-moment argument pins the entire vector $q = 0$: a weak dominator of *any* exact-size test must match its rejection mass and all three first moments under P_0. The usual dominators (transfers, gates) all fail this matching, which is why they do not exist here.

Third, every nonnegative direction is either feasible or a negated null direction. If $a \ge 0$ is bad, then $a_3 > \max\{a_1, a_2\} \ge \min\{a_1, a_2\}$, so $-a \in \Lambda_0$ (indeed $-a_3 \le -\min\{a_1, a_2\} = \max\{-a_1, -a_2\}$). In the Bayes-representation heuristic for complete classes, a facet of an acceptance region can be supported either by alternative prior mass along its normal or by null prior mass along its negated normal. Since $\overline{\Lambda}_1 \cup (-\Lambda_0) \supseteq \mathbf{R}^3_+$, *every* facet orientation of a lower convex region is supportable from one side or the other. There is no orientation-based obstruction to being a limit of Bayes tests, hence no cheap route to inadmissibility.

The upshot: if an inadmissible nice test exists, the phenomenon must be global and second-order, not directional. The next two results make this precise by carving out large admissible classes.

2 Feasibly supported regions are admissible

Say A is *feasibly supported* if

$$A = \bigcap_{a \in F} \{x : a'x \le h_A(a)\}, \qquad F := \{a \text{ feasible} : h_A(a) < \infty\},$$

that is, the supporting halfspaces with feasible normals already determine A. For a polyhedron this holds if and only if every facet normal is feasible.

Proposition 1 (Feasible support suffices)**.** *Every feasibly supported nice test is α-admissible against all level-α tests, including randomized competitors.*

Proof. Let ψ weakly dominate ϕ_A and $h = \psi - \phi_A$, so $\mathrm{E}_0 h = 0$, $\{h > 0\} \subseteq A$, $\{h < 0\} \subseteq A^c$. Fix a nonzero $a \in F$; then $a \in \Lambda_1$, so dominance along the ray ta, $t > 0$, gives $\mathrm{E}_0[e^{ta'X}h] \ge 0$. Since h^+ is supported on $A \subseteq \{a'x \le h_A(a)\}$, the tilt bound of the rectangle proof applies verbatim: if $\mathrm{E}_0[h^-1\{a'X \ge h_A(a) + \varepsilon\}] > 0$ for some $\varepsilon > 0$, then $\mathrm{E}_0[e^{ta'X}h] \le e^{th_A(a)}\mathrm{E}_0 h^+ - e^{t(h_A(a)+\varepsilon)}\mathrm{E}_0[h^-1\{a'X \ge h_A(a)+\varepsilon\}] < 0$ for large t, a contradiction. Hence $h^- = 0$ a.s. on $\{a'x > h_A(a)\}$. The complement $A^c = \bigcup_{a \in F}\{a'x > h_A(a)\}$ is open, so by the Lindelöf property a countable subfamily covers it, and therefore $h^- = 0$ a.s. on A^c. Since $\{h^- > 0\} \subseteq A^c$, this gives $h^- = 0$ a.s.; then $\mathrm{E}_0 h = 0$ forces $\mathrm{E}_0 h^+ = 0$, so $h = 0$ P_0-a.s., and mutual absolute continuity of Gaussian shifts finishes the proof as in the main note. □

Note that the proof uses only the confinement step; the first-moment identity is not needed. Two remarks. First, this is strictly weaker as a hypothesis than the feasible-normal condition of the extension plan, which demands that *every* finite-support direction be feasible. Only the directions needed to *define* A must be feasible; sums of facet normals that leave the feasible set, such as $(1, 1, 2)$, are irrelevant. Second, it settles a question the main note left open:

Corollary 1 (The symmetric gate is admissible)**.** *The symmetric gated region $A_b^{\mathrm{g}} = \{x_1 \le \kappa,\ x_2 \le \kappa,\ x_1 + x_3 \le b,\ x_2 + x_3 \le b\}$ has facet normals e_1, e_2, $(1, 0, 1)$, $(0, 1, 1)$, all feasible, and is therefore admissible — even though its finite-support direction $(1, 1, 2)$ is infeasible and the extension-plan condition fails.*

So the symmetric gate and the rectangle are *both* admissible and neither dominates the other, consistent with the power comparison in Section 2 of the main note.

3 One bad halfspace can be added for free

Proposition 2 (Feasible support plus one halfspace)**.** *Let $A = A_g \cap \{x : u'x \le c\}$ be nice, where A_g is an intersection of halfspaces with feasible normals and $u \ne 0$ is arbitrary. Then ϕ_A is α-admissible.*

Proof. As above, $\mathrm{E}_0 h = 0$ and now also $q = \mathrm{E}_0[Xh] = 0$, so in particular $\mathrm{E}_0[u'Xh] = 0$. The confinement step of Proposition 1 gives $h^- = 0$ a.s. on A_g^c, so h^- is supported on $A^c \cap A_g = \{u'x > c\} \cap A_g$, while h^+ is supported on $A \subseteq \{u'x \le c\}$. With $M := \mathrm{E}_0 h^+ = \mathrm{E}_0 h^-$,

if $M > 0$ then $\mathrm{E}_0[u'Xh^+] \le cM$ and $\mathrm{E}_0[u'Xh^-] > cM$ (strictly, since $u'X > c$ a.s. on $\{h^- > 0\}$ and $P_0(u'X = c) = 0$), whence $\mathrm{E}_0[u'Xh] < 0$, contradicting $u'q = 0$. So $M = 0$, $h = 0$ P_0-a.s., and the conclusion follows as before. □

This is the rectangle proof in its natural generality: the rectangle is $\{x_1 \le c_1, x_2 \le c_2\}$ (feasible normals e_1, e_2) intersected with the single bad halfspace $\{x_3 \le c_3\}$. Two further corollaries are worth recording.

Corollary 2. *The redundant-only halfspace $\phi_3 = I\{x_3 > z_\alpha\}$ is admissible for every $\alpha \in (0,1)$: here $A_g = \mathbf{R}^3$ and $u = e_3$, so $q_3 = 0$ alone yields the contradiction. This removes the Neyman–Pearson step from the proof of Proposition 2 of the main note, and with it the $\alpha < 1/2$ restriction flagged in my earlier comments.*

Corollary 3. *The following exact-size regions are admissible: (i) any rectangle with one additional* feasible *cut, e.g. $\{x_1 \le \kappa_1,\, x_2 \le \kappa_2,\, x_3 \le \kappa_3,\, a'x \le c\}$ with a feasible (here A_g collects the core faces and the cut, and the sole bad halfspace is the x_3 face); and (ii) any region of the form $\{x_1 \le \kappa_1,\, x_2 \le \kappa_2,\, a'x \le c\}$ with a bad — the two core faces plus a single bad cut, with no x_3 face.*

Remark 1 (A rectangle plus a bad cut is *not* covered). Proposition 2 does not apply to $\{x_1 \le \kappa_1,\, x_2 \le \kappa_2,\, x_3 \le \kappa_3,\, a'x \le c\}$ with a bad and the cut binding: the x_3 face is itself a bad halfspace (e_3 is infeasible), so this region has *two* exposed bad facets and exceeds the "one arbitrary halfspace" budget. Nor can the proof be repaired by a cleverer choice of u: any separating functional ℓ must have $\ell'x$ bounded below on the gap region, which contains points with $x_3 = \kappa_3 + \varepsilon$ and $x_1, x_2 \to -\infty$; this forces $\ell_1 = \ell_2 = 0$, i.e. $\ell \propto e_3$. But when the cut binds ($c < a_1\kappa_1 + a_2\kappa_2 + a_3\kappa_3$), the lobe $\{a'x > c\} \cap A_g$ contains points with $x_3 < \kappa_3 = \sup_A x_3$, so e_3 fails as well. No separating functional exists, and this region joins the frontier of Section 4.

4 Where a counterexample must live

Combining the two propositions: an inadmissible nice test must have at least *two* exposed halfspaces with bad normals, irreducibly — the region cannot be rewritten as feasible halfspaces plus a single extra one. (A feasibly supported set cannot have a bad facet normal, since facet normals of an intersection of halfspaces lie in the closed cone of the normals used.) Moreover, the two bad facets must be arranged so that no single linear functional separates A from the gap region $A_g \setminus A$; otherwise the proof of Proposition 2 goes through with that functional in place of u.

Two candidate families sit at this frontier. The simplest is the rectangle with one bad cut of Remark 1,

$$A^{\mathrm{cut}} := \{x_1 \le \kappa_1,\ x_2 \le \kappa_2,\ x_3 \le \kappa_3,\ a'x \le c\}, \qquad a \text{ bad, cut binding,} \tag{1}$$

whose two bad facets are e_3 and a. The second is

$$A(\lambda) := \{x_1 \le \kappa,\ x_2 \le \kappa,\ x_1 + \lambda x_3 \le b,\ x_2 + \lambda x_3 \le b\}, \qquad \lambda > 1, \tag{2}$$

with (κ, b) calibrated to exact size; the gate normals $(1, 0, \lambda)$ and $(0, 1, \lambda)$ are bad, both facets are exposed, and the gap region has two lobes, one along each bad facet, which no single linear functional can dominate simultaneously (the argument of Remark 1 adapts). Every tool above fails on both families, and I believe this is the true frontier: resolving admissibility of, say, $A(2)$ or of A^{cut} with $a = (1, 1, 3)$ would either produce the desired example or strongly indicate that none exists.

One observation tempers optimism about finding inadmissibility here. Both families deform continuously into admissible tests. At $\lambda = 1$, $A(\lambda)$ is the symmetric gate — admissible by Proposition 1; as $\lambda \to \infty$ with b rescaled, it converges to a rectangle — admissible by Proposition 2. Likewise A^{cut} tends to the plain rectangle as the cut is relaxed (c increased until it stops binding) and to a region covered by Corollary 3(ii) as the x_3 face is relaxed ($\kappa_3 \to \infty$). Inadmissibility, if present, would have to appear and then disappear along these paths. Possible, but it would be a delicate phenomenon.

5 Conjecture and two ways to attack it

Conjecture. In the two-core model, every closed, convex, coordinatewise-lower exact-size test is admissible.

My reasons for leaning this way: the three structural facts of Section 1 (no disqualifying directions; all first moments pinned; every facet orientation Bayes-supportable), plus the fact that every case resolved so far — all nonnegative halfspaces, all finite rectangles, ϕ_3, the symmetric gate, the regions of Corollary 3 — has come out admissible, plus the interpolation observation above. But the conjecture is genuinely open on the families (1) and (2), and I would not bet heavily.

Attack 1 (hunting: a linear-programming search). Inadmissibility of ϕ_A is the feasibility of a linear program in the perturbation $h = \psi - \phi_A$: maximize $\sum_k \mathrm{E}_{\mu_k^*} h$ over a few probe alternatives μ_k^*, subject to $-\phi_A \le h \le 1 - \phi_A$, $\mathrm{E}_\mu h \ge 0$ for μ in a fine net of Λ_1 (include rays of large norm — these encode the tilt constraints that do the confining), and $\mathrm{E}_\mu(\phi_A + h) \le \alpha$ for μ in a net of Λ_0. Discretize x on a truncated grid, impose the $1 \leftrightarrow 2$ symmetry to halve the problem, and solve for $A(2)$. A robustly positive value as the nets refine yields a concrete dominator candidate to verify analytically; a value collapsing to zero is strong evidence for admissibility, and the dual solution — weights on the binding μ's — reads off the prior sequence for Attack 2. Either outcome is informative.

Attack 2 (proving: a Bayes representation). To prove admissibility of $A(\lambda)$ directly, exhibit it as a suitable limit of Bayes tests and apply a Blyth-type argument. The structure of Section 1 says which priors to try: alternative prior mass escaping to infinity along the feasible facet normals e_1, e_2 (these generate the core faces), and null prior mass escaping along $-(1, 0, \lambda)$ and $-(0, 1, \lambda)$ — both in Λ_0 since $\lambda > 0$ — to generate the bad facets from the null side, with weights tuned so the limiting acceptance region is the polyhedron $A(\lambda)$. The delicate point is that the null constraints far along those rays are slack, so the weights must grow at the right exponential rate; whether this can be done consistently for two bad facets at once is exactly the same question the LP probes numerically.

Literature worth revisiting. The classical admissibility theory for Gaussian means — Birnbaum's convexity theorem, Stein's necessary condition, Matthes–Truax for exponential families, and the complete-class and one-sided-testing results of Marden, Perlman, and Brown–Marden — is where both the necessary-condition machinery and the limit-of-Bayes sufficiency arguments live. As noted in Section 1, the necessary conditions are vacuous in this model, which is itself the clearest signal that the two-core problem sits in the regime where those papers' sufficiency techniques, not their necessity techniques, are the relevant ones. (Citations from memory; worth checking exact statements before relying on them.)